

Programme: B E – Information Science & Engineering
Internal Assessment – I

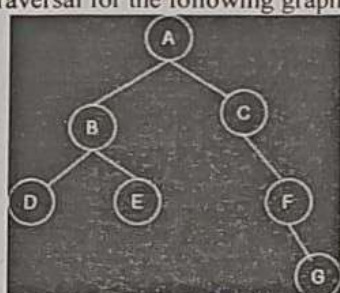
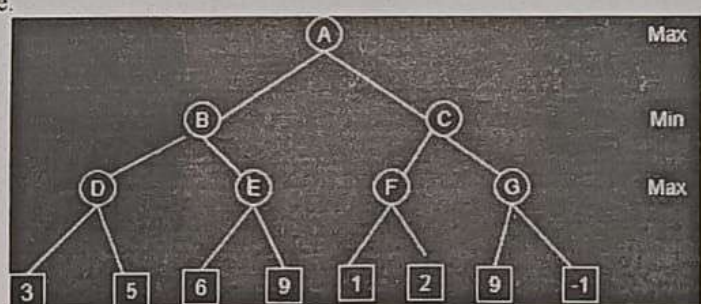
TERM : 10-09-2025 - 31.12.2025	COURSE NAME: Artificial Intelligence
DATE : 29-10-2025 TIME: 11:00 to 12:00 pm	COURSE CODE : IS52
MAX MARKS: 30	PORTIONS : L1 – L15



Mobile Phones are banned

Instructions to Candidates: Answer any TWO full questions.

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)	CO	Marks																		
1.a	Explain the four approaches used to describe AI systems: thinking humanly, acting humanly, thinking rationally, and acting rationally with a suitable example.	L2	CO1	06																		
b	Explain how Bidirectional Search is efficient compare to BFS and DFS methods. Show the DFS and BFS traversal for the following graph to reach the goal state G. 	L3	CO2	06																		
c	Apply minmax method for the following game tree to compute min-max values at each node. 	L3	CO3	03																		
2.a	Describe the structure and functioning of a Model-Based Reflex Agent in Artificial Intelligence with a suitable diagram and pseudocode.	L2	CO1	06																		
b	Solve the following 8-puzzle problem using A* search algorithm. Given start state and goal state. Find the most cost-effective path to reach the goal state from start state (consider $g(n)$ = depth of node and $h(n)$ = number of misplaced tiles). <table border="1" data-bbox="386 1926 643 2065"><tr><td>1</td><td></td><td>3</td></tr><tr><td>4</td><td>2</td><td>6</td></tr><tr><td>7</td><td>5</td><td>8</td></tr></table> Start state <table border="1" data-bbox="807 1926 1062 2065"><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td>5</td><td>6</td></tr><tr><td>7</td><td>8</td><td></td></tr></table> Goal State	1		3	4	2	6	7	5	8	1	2	3	4	5	6	7	8		L3	CO2	06
1		3																				
4	2	6																				
7	5	8																				
1	2	3																				
4	5	6																				
7	8																					

c	How does Alpha-Beta pruning improve efficiency Compare to Minimax algorithm?	L2	CO3	03
3.a	For each of the following intelligent systems, define the PEAS (Performance measure, Environment, Actuators, Sensors) description of the task environment: i. Smart traffic control system ii. Intelligent medical diagnosis system	L3	CO1	06
b	Why is uninformed search known as blind search? Explain uniform cost search with a suitable example.	L3	CO2	06
c	Define the terms alpha and beta in Alpha-Beta pruning and state their roles.	L2	CO3	03

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create

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Internal Assessment – II

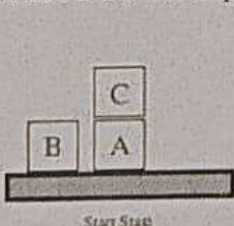
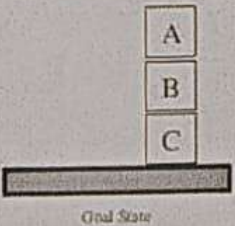
TERM : 10-09-2025 - 31.12.2025	COURSE NAME: Artificial Intelligence
DATE : 23-12-2025 TIME: 3:00 to 4:00	COURSE CODE : 23IS52
MAX MARKS: 30	PORTIONS : L16 – L28



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Instructions to Candidates: Answer any TWO full questions.

Marks: 15x2=30

Q. NO	Questions	Blooms Levels (L1 to L6)*	CO	Marks
1.a	Convert the following sentences into First-Order Logic i. If a square contains a pit, then all adjacent squares are breezy. ii. If there is glitter in a square, then that square contains gold. iii. There is gold in square (x, y)	L3	CO3	03
b	"As per the law, it is a crime for an American to sell weapons to hostile nations. Country A, an enemy of America, has some missiles, and all the missiles were sold to it by west, who is an American citizen." Solve the above problem using forward chaining method and prove that "west is criminal". Mention all the steps involved.	L4	CO4	06
c	Justify the statement "Can Machines Really Think?" with respect to "The Chinese room" and "Consciousness and qualia" philosophical arguments.	L4	CO5	06
2.a	Explain the connectives used in propositional logic with suitable examples.	L2	CO3	03
b	Describe the steps for converting First Order Logic sentences to clause form and demonstrate resolution for the following example. Example: • All humans are mortal • Socrates is a human Prove Socrates is mortal	L4	CO4	06
c	Discuss the AI ethical concerns related to surveillance, job displacement, misinformation, and digital inequality.	L2	CO5	06
3.a	Given that square (1,2) is breezy and square (2,1) is not breezy, what can the agent infer about the presence of pits? Explain.	L2	CO3	03
b	Explain with examples the purpose of unification and lifting operations in first-order logic.	L2	CO4	06
c	Write PDDL description to solve the following 'Blocks world problem'.  Start State  Goal State	L3	CO5	06

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create